THILAKSHANA RAJAPAKSE

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PROFESSIONAL SUMMARY

Analytical and detail-oriented Data Science graduate (BSc) specializing in Data Engineering, data warehousing, and relational/NoSQL database management (MySQL, Oracle DB, MongoDB Cloud). Technically proficient in Python, Java, and JavaScript, with hands-on experience in data mining, statistical modeling, and IoT integrations. Adept at structuring complex datasets for analytical use, version controlling via GitHub, and collaborating across cross-functional teams to solve architectural data challenges.

WORK EXPERIENCE

Data Analytics Intern
Sandbox – Sri Jayawardenepura Kotte, Sri Lanka (On-site)
February 2026 – Present

- Performed data sampling, regression analysis, and statistical modeling to identify market trends and support data-driven business decisions.
- Collected, mined, and analyzed large datasets to generate interactive visual analytics, dashboards, and reporting frameworks for stakeholders.
- Contributed to location intelligence and spatial market analysis initiatives to uncover growth opportunities and potential business expansions.
- Gathered, documented, and translated complex business and user requirements into technical specifications for software development initiatives.
- Conducted rigorous system testing, data validation, and quality assurance troubleshooting to ensure software solutions met strict business requirements.
- Analyzed qualitative and quantitative research data regarding workforce trends and organizational culture to deliver comprehensive insight reports and strategic recommendations.

AI / ML Engineer Intern
OXZON AI – Northampton, United Kingdom (Remote)
October 2024 – April 2025

- Developed, tested, and implemented predictive machine learning models in Python for a variety of data-driven software applications.
- Executed advanced data preprocessing, data cleaning, and feature engineering to optimize model reliability, accuracy, and overall performance.
- Evaluated model performance through rigorous testing and iteration, adjusting hyper-parameters to enhance algorithmic results.
- Collaborated effectively within a remote, Agile team environment to design, document, and deploy scalable AI-based technical solutions.

TECHNICAL SKILLS

- **Data Engineering & Analysis:** Data Mining, Data Engineering, Data Modeling, Data Warehousing, Statistical Modeling, Regression Analysis, Visual Analytics, UX Design
- **Programming Languages:** Python, Java, JavaScript, R (R Studio)
- **Databases & Cloud:** MySQL, SQLite, Oracle DB, MongoDB Cloud
- **Tools & Frameworks:** GitHub, Flask, Power BI, MS Excel, Node-RED, MQTT, Arduino IDE, AWS (Free Tier)
- **Languages:** English, Sinhala
- **Soft Skills:** Problem Solving, Technical Presentation, Team Collaboration

PROJECTS

MindTrack360 — Bipolar Risk Detection & Emotion Tracking System
Technologies: Flask, MongoDB, DistilBERT, Flan-T5, SHAP explainer, VS Code, Python
November 2024 – October 2025

- Developed a web-based system to assess bipolar disorder risk using social media data and behavioral patterns.
- Designed a multi-component architecture incorporating linguistic analysis with explainable AI (PsyTex), chatbot-based emotion tracking (EmoTra), and personalized behavioral recommendations (BeWell AI).
- Applied machine learning and natural language processing (NLP) techniques to analyze user-generated text data, mapping emotional patterns to identify potential risk levels.
- Published a dedicated research paper detailing the project framework in IEEE Xplore (ICAC Conference). URL: https://ieeexplore.ieee.org/document/11361478

EcoSphere — Smart Home Climate Management System
Technologies: Arduino IDE, MongoDB, DHT22 & DHT11 Sensors, Servo Motor (SG90), ESP32 Microcontroller, Node-RED, MQTT
February 2025 – May 2025

- Engineered an autonomous climate regulation system utilizing DHT11/DHT22 sensors to monitor real-time environmental data and automatically adjust windows via servo motors.
- Built a dynamic Node-RED data visualization dashboard utilizing an ARIMA model to forecast local climate trends for upcoming 12-month periods.
- Architected communication protocols over ESP32 microcontrollers utilizing MQTT to facilitate low-latency communication between hardware sensors, actuators, and the central system dashboard.

Airline Customer Satisfaction Classifier
Technologies: Python, Flask, Plotly Dash, AWS Free Tier, Classification Models
February 2024 – May 2024

- Developed a predictive classification system to evaluate customer satisfaction metrics and help airlines optimize operational frameworks.
- Utilized machine learning classification models to isolate root service deficiencies, analyze passenger behavior trends, and predict satisfaction scores.
- Deployed the data analytics model through a web interface to provide marketing and promotion optimization insights based on consumer segmentations.

Web System for "Suwa Arana" Ayurvedic Treatment Center
Technologies: MERN Stack (MongoDB, Express.js, React.js, Node.js)
June 2023 – October 2023

- **End-to-End Digitization:** Developed a comprehensive, full-stack web application that completely digitized and automated the manual workflows of an active treatment center, replacing physical paper records with a centralized database system.
- **Modular Micro-Architecture:** Engineered a scalable, 7-component modular system within a collaborative team environment, successfully integrating core modules including User Authentication, Secure Payment Gateways, Doctor Channeling, Appointment Scheduling, and Inventory/Stock Management.
- **Data Modeling & Optimization:** Designed and optimized the MongoDB database schemas to manage complex data relationships between multiple user roles (patients, doctors, administrators), ensuring real-time synchronization, secure data access control, and low-latency query handling.

EDUCATION

BSc (Hons) in Information Technology — Specializing in Data Science
Sri Lanka Institute of Information Technology (SLIIT)
2022 – 2025 | CGPA: 3.5/4.0 | **Merit Award for Academic Excellence - March 2026**

REFERENCES

Ms. Malithi Nawarathne
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